Figure identity

Colour-blind-safe identity for all papers: Okabe-Ito categorical colours, batlow/vik for ordered data, Liberation Mono, SVG + PDF export.

1 Categorical colours - fixed order, colour follows the entity

#	Mark	Name	Fill	Edge	White	Normal	Deutan	Protan	Tritan	Grey
1	● ○	**blue**	#005B90	#013455	7.2					
2	■ □	**vermillion**	#BB4000	#7C2800	5.5					
3	▲ △	**bluish green**	#009E73	#066F50	3.4					
4	◆ ◇	**orange**	#E69F00	#AE7809	2.3					
5	▼ ▽	**sky blue**	#56B4E9	#2288BB	2.3					
6	⬟ ⬠	**reddish purple**	#CC79A7	#9E4F7C	3.1					

White = contrast against white (≥ 3 is the ACS rule for graphics; red values rely on the dark edge + marker shape). Slots 1-3 also differ in brightness (1.33-2.12:1 between any two), so they stay apart when hue is lost. Worst separation within the first k slots, OKLab ΔE×100 under deutan/protan simulation (Machado 2009): k = 2: 18.7, k = 3-5: 11.2, k = 6: 7.5. Six series need distinct marker shapes; more than six: split the panel or use a ramp. Alternatives: ff.use(palette='B' or 'C').

2 Neutrals and reserved colours

black	**grey_dk**	**grey**	**grey_lt**	**grid**	**grid_minor**	**yellow**
#000000	#404040	#808080	#B3B3B3	#E6E6E6	#F2F2F2	#F0E442
text, axes, model	2nd model, notes	literature data	guides: y=x, 0	major grid, bands	minor grid	areas only

3 Ordered data - sequential and diverging (Crameri colour maps)

batlow
ff.cmap('seq')

Deutan
Protan
Tritan
Grey

ff.ramp(3) #011959 #477150 #E9995C

ff.ramp(5) #011959 #165062 #477150 #95872C #E9995C

for isotherms, isobars, compositions: dark = low

vik
ff.cmap('div'), centre 0

Deutan
Protan
Tritan
Grey

Never jet/rainbow: uneven lightness invents features and fails with colour-vision deficiency and in greyscale.

4 Why these choices

- Okabe-Ito (Okabe & Ito 2008; Wong, Nat. Methods 8, 441, 2011) is the accessible palette cited by the Nature figure guide; about 8% of men of N-European ancestry have red-green colour-vision deficiency.
- Slot order computed, not guessed: it maximises the worst-case CVD separation while slots 1-3 keep >= 3:1 contrast on white. Blue and vermillion are darker than published Okabe-Ito (#0072B2, #D55E00): a colour-blind reader found vermillion and green hard to tell apart at equal brightness (1.13:1; now 1.60:1, confirmed effortless).
- Liberation Mono (free, SIL OFL, bundled) keeps the BeautifulFigures monospace look with Courier's letter widths. Elsevier names Courier among its standard fonts; ACS recommends Arial/Helvetica but only sets a 4.5 pt minimum; Nature requires sans-serif (ff.use('nature', font='Arial')).
- Faint grid and outward ticks come from BeautifulFigures; grid lines are 0.5 pt (ACS minimum line weight) in very light grey so they stay under the data.
- batlow/vik (Crameri, Shephard & Heron, Nat. Commun. 11, 5444, 2020) are perceptually uniform and stay readable with CVD and in greyscale.
- Two-tone markers (fill + darker edge of the same hue) keep light hues visible on white.

Set in Liberation Mono. Install the bundled TTFs (fairfig/fonts) on Windows to edit the SVGs with the same face.

Figure identity

Colour never works alone: every series also has a marker shape or line style, so figures survive greyscale print and colour-vision deficiency.

5 Marks

Experiment	ff.points(i)	filled: liquid x1, phase I
2nd phase	ff.points(i, hollow=True)	open: vapour y1, phase II
Literature	ff.points(i, lit=True)	grey, under your data
Model	ff.curve(i, style=s)	1 model: series colour several: black, styles
Guides	ff.diagonal, ff.hline, ff.band	y=x 0 ±u 2-phase

6 Type and axis labels

Liberation Mono, one size for all text (8 pt ACS/Elsevier, 7 pt Nature preset); hierarchy by weight: panel labels bold, 1 pt larger. Quantities italic, units upright, IUPAC T/K (units='paren' gives T (K)); unit factors joined by a middle dot. Black text only.

T/K	ff.label("T", "K")	ρ/(kg·m^{-3})	ff.label(r"\rho", "kg m^-3")
p/kPa	ff.label("p", "kPa")	u/(m·s^{-1})	ff.label("u", "m s^-1")
x_1, y_1	ff.label("x_1, y_1")	η/(mPa·s)	ff.label(r"\eta", "mPa s")

a **b** **c** ff.panel_labels(axes): above the frame, top-left (Elsevier preset: (a), (b))

7 Layout and export - draw at final size, never rescale

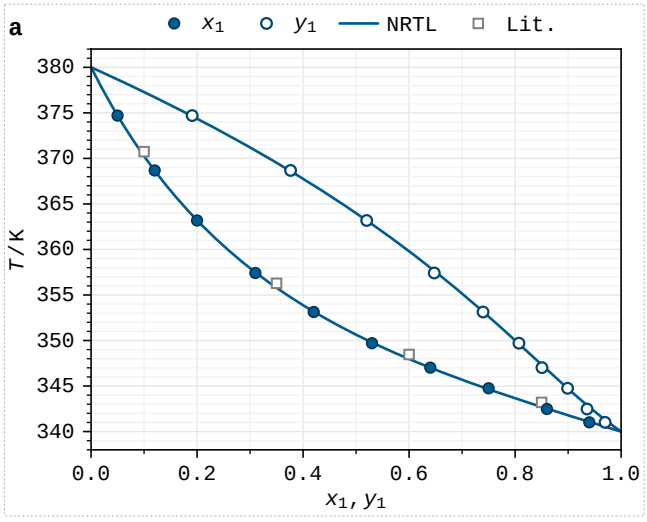
preset	single	1.5 col	double	text	max h
use("acs")	84.6	127	177.8	8 pt	233
use("elsevier")	90.0	140	190.0	8 pt	-
use("nature")	89.0	120	183.0	7 pt	170

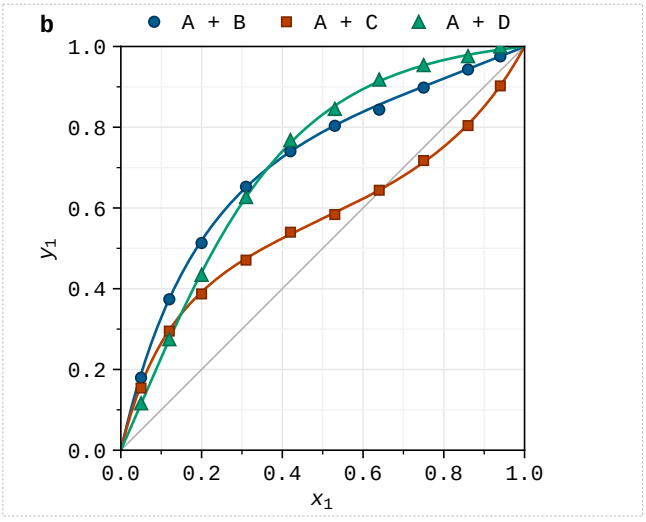
widths and heights in mm

closed box, ticks out (left/bottom)
faint grid: major + minor, 0.5 pt
legend on top: ff.legend(ax)
lines 1.0 pt, axes 0.6 pt
markers 4 pt, edge 0.6 pt
export: SVG (editable) + PDF

Submit the PDF (ACS and Elsevier do not take SVG); keep the SVG for editing in a vector editor.

8 At 1:1 - two ACS single-column figures (84.6 x 67.7 mm)





9 Workflow

```
import fairfig as ff; ff.use("acs")      # once per notebook
fig, ax = ff.figure("single")           # or ff.subplots(1, 2)
ax.plot(x, T, **ff.points(0)); ax.plot(xm, Tm, **ff.curve(0))
ax.set(xlabel=ff.label("x_1"), ylabel=ff.label("T", "K")); ff.legend(ax)
ff.save(fig, "figures/fig1")            # -> fig1.svg + fig1.pdf
```